

Patrick Chen

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EDUCATION

Bachelor of Software Engineering (Honors & Co-op) – GPA: 3.9 2025 – Present
*University of Waterloo | **Earned Governor General's Academic Medal** | SE'30 Yearbook Rep* Waterloo, ON

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, SQL, JavaScript/TypeScript, HTML/CSS
Frameworks & Libraries: Flask, Django, FastAPI, React.js, SQLAlchemy, NumPy, Pandas, scikit-learn, Matplotlib, OpenCV, Supabase, Firebase
Developer Tools: Git, GitLab CI/CD, GitHub Actions, Docker, Linux, Bash, Postman, Figma, Adobe Photoshop

EXPERIENCE

Software Engineer Jan 2026 – Present
UW Formula Electric Waterloo, ON

- Develop **embedded telemetry software** for a 600V Formula race car, processing 3+ real-time vehicle signals through **CAN communication pipelines**
- Built internal data logging and visualization tooling analyzing battery temperature, voltage, and motor efficiency metrics, **reducing debugging time** during testing cycles by **roughly 30%**

Software Engineer Intern Jun 2024 – Aug 2024
Simply Embedded Calgary, AB

- Engineered a CRUD **Django-based manufacturing platform** integrating Odoo ERP with IndaBOM to automate BOM workflows and synchronize **more than 1,200 inventory records** across production systems
- Designed **backend API services** enabling reliable communication between ERP and internal databases, **reducing manual BOM processing time by approximately 70%**
- Implemented **QR and barcode scanning workflows** to process manufacturing scans with validation safeguards

Engineering & Test Intern Jun 2023 – Aug 2023
Simply Embedded Calgary, AB

- Developed an internal Odoo application using **Python and XML** to track vendor pricing across more than 250 electronic components used in embedded hardware production
- Integrated the **Octopart REST API** to automatically retrieve supplier pricing from over 20 vendors and generate structured CSV reports containing **200–500 parts** per export
- Performed **regression testing** on 10+ embedded devices with **binary-search boundary testing** techniques

Co-Founder Jan 2025 – Present
CodeUnity Calgary, AB

- Co-founded a nonprofit delivering programming education to **over 100 students** through weekly courses covering Python, C++, object-oriented programming, and algorithm
- Designed a structured curriculum with over **10 project-based modules**, building games & automation tools

PROJECTS

AlgoLize — Graph/Tree Algorithm Visualizer   — React, TypeScript, Vite Feb 2026

- Built an interactive algorithm visualization platform enabling step-by-step debugging of graph algorithms using an integrated **Monaco editor** and **SVG rendering engine**
- Implemented a runtime visualization engine supporting 7 event types and timeline playback for **BFS, DFS, and shortest-path algorithms** across graphs with **more than 1,000 nodes**

Guardify — Scam Detection Chrome Extension  — React, FastAPI, Python Jan 2026

- Developed a Chrome extension detecting fraudulent Shopify listings using **machine learning risk scoring**
- Engineered a **FastAPI backend** extracting more than 30 behavioral features and returning fraud predictions with **80% classification accuracy**

ParsAR — AR Social Practice Mobile App — UofTHacks 13  — Unity, C#, AR Foundation Jan 2026

- Developed an **AR application** enabling AI-driven social interaction practice within **48-hour hackathon**
- Integrated **ElevenLabs API** for real-time text-to-speech & **Google Cloud NLP** for conversational interaction

Pac-Man Game  — Java, TypeScript, HTML5 Canvas Feb 2026

- Developed a browser-based Pac-Man clone implementing deterministic grid movement, collision detection, and a **real-time 60 FPS game loop**
- Implemented **ghost AI using BFS pathfinding** across a 19x21 tile map with predictive targeting behavior inspired by the original arcade mechanics